

## CE 452: Open Channel Flow

Course Code: CE 452

Course Name: Open Channel Flow

L-T-P-C: 2-1-0-3

Prerequisites: CE 251 - Hydraulics Engineering

Intended for: UG (Civil Engineering)

Mutual Exclusion: None

Approval: 61st BoA

### Preamble

This course provides a comprehensive understanding of open channel flow and hydraulic systems, covering fundamental concepts, flow resistance, energy and momentum principles, uniform and non-uniform flow, canal design methodologies, unsteady flow phenomena, pipe flow analysis, and hydraulic modeling techniques. The course emphasizes both theoretical foundations and practical applications in water resources engineering, including design of irrigation canals, analysis of hydraulic structures, and modeling of complex flow systems. Students will develop skills in analyzing and designing various hydraulic systems encountered in civil and environmental engineering practice.

### Course Modules

- **Module 1: Introduction to Open Channel Flow:** Open Channel Flow Definition, Types of Channels, Geometric parameters of a channel, Classification of Open Channel Flow, Continuity equation for open channel flow, Momentum equation and its applications in open channel flow. (6 Hours)
- **Module 2: Uniform Flow:** Resistance flow formula, Velocity distribution in open channels, Equivalent roughness coefficient, Velocity coefficients, Uniform flow in rigid boundary channel, Uniform flow in mobile boundary channel. Concept of Specific Energy, Critical depth and its computation, Alternate depth, Choke condition, Specific Force diagram, Sequent depth and its applications. (8 Hours)
- **Module 3: Gradually Varied Flow:** Governing equation of Gradually Varied Flow (GVF), Dynamic equation of GVF, Classification of Gradually Varied Flow (types of flow profiles - M, S, C, H, A curves), Computation of GVF profile (direct step method, standard step method). (8 Hours)
- **Module 4: Rapidly Varied Flow:** Rapidly Varied Flow characteristics, Hydraulic Jump (formation, types, energy loss, length, and location), Flow over a Hump (free flow and submerged flow conditions), Flow in Channel Transition (expansions and contractions, design considerations). (8 Hours)
- **Module 5: Unsteady Flow:** Occurrence of Unsteady Flow, Waves and their classification (gravity waves, dynamic waves), Celerity of wave in open channels, Surges (positive and negative surges), Governing equation for unsteady flows, St. Venant Equations, Boussinesq Equations. (6 Hours)
- **Module 6: Numerical Methods:** Method of Characteristics, Initial and Boundary Conditions, Characteristic Grid Method, Method of Specified Intervals, Introduction to Finite Difference Methods, Flood Modeling using HEC-RAS. (6 Hours)

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### Tutorial Modules

Problem-solving sessions covering numerical applications of open channel flow equations; Computation of flow profiles; Case studies on real-world hydraulic engineering projects; Computer-aided analysis of hydraulic systems.

### Text Books

1. Subramanya, K., *Flow in Open Channels*, 4th Edition, McGraw Hill Education, India, 2021
2. Chaudhry, M. H., *Open Channel Flow*, 2nd Edition, Springer, USA, 2008

### References

1. Chow, V. T., *Open Channel Hydraulics*, McGraw Hill, USA, 1959
2. French, R. H., *Open Channel Hydraulics*, McGraw Hill, USA, 1985
3. Chanson, H., *The Hydraulics of Open Channel Flow: An Introduction*, 2nd Edition, Elsevier, UK, 2004
4. Henderson, F. M., *Open Channel Flow*, Macmillan, USA, 1966
5. Modi, P. N. and Seth, S. M., *Hydraulics and Fluid Mechanics*, Standard Book House, India, 2020
6. Streeter, V. L. and Wylie, E. B., *Fluid Mechanics*, McGraw Hill, USA, 1985